1. What is the rank of an m x n zero matrix?
- **(A)** 0
- **(B)** 1
- **(C)** min(m, n)
- **(D)** m * n

2. An elementary row operation on matrix A does not alter its:
- **(A)** Eigenvalues
- **(B)** Rank
- **(C)** Trace
- **(D)** Determinant

3. A square matrix A is called singular if and only if:
- **(A)** Trace(A) = 0
- **(B)** det(A) = 0
- **(C)** Rank(A) = n
- **(D)** A = A^T

4. The transpose of a product of two matrices (A * B)^T is equal to:
- **(A)** A^T * B^T
- **(B)** B^T * A^T
- **(C)** (B * A)^T
- **(D)** A * B

5. For any real square matrix A, the matrix A + A^T is always:
- **(A)** Skew-symmetric
- **(B)** Symmetric
- **(C)** Orthogonal
- **(D)** Unitary

6. For any real square matrix A, the matrix A - A^T is always:
- **(A)** Symmetric
- **(B)** Skew-symmetric
- **(C)** Identity
- **(D)** Diagonal

7. The diagonal elements of a skew-symmetric matrix are always:
- **(A)** All ones
- **(B)** Purely imaginary
- **(C)** All zeros
- **(D)** Real non-zero

8. The determinant of an orthogonal matrix Q (Q^T * Q = I) is always:
- **(A)** 0
- **(B)** 1 only
- **(C)** +1 or -1
- **(D)** Any positive real

9. In Row Echelon Form (REF), the leading entry in each non-zero row is called a:
- **(A)** Kernel
- **(B)** Pivot
- **(C)** Eigenvalue
- **(D)** Residue

10. The rank of an m x n matrix A is bounded above by:
- **(A)** m + n
- **(B)** min(m, n)
- **(C)** max(m, n)
- **(D)** m * n

11. If matrix A is of order 3x3 and det(A) = 5, then det(2*A) equals:
- **(A)** 10
- **(B)** 30
- **(C)** 40
- **(D)** 8

12. If A is an invertible matrix of order n, then det(adj(A)) equals:
- **(A)** det(A)^n
- **(B)** det(A)^(n-1)
- **(C)** det(A)^(n+1)
- **(D)** 1 / det(A)

13. The sum of the eigenvalues of any square matrix A equals its:
- **(A)** Determinant
- **(B)** Rank
- **(C)** Trace
- **(D)** Nullity

14. The product of the eigenvalues of any square matrix A equals its:
- **(A)** Trace
- **(B)** Determinant
- **(C)** Rank
- **(D)** Norm

15. If lambda is an eigenvalue of an invertible matrix A, then an eigenvalue of A^-1 is:

(A) -lambda
(B) 1 / lambda
(C) lambda^2
(D) sqrt(lambda)

16. If lambda is an eigenvalue of matrix A, then an eigenvalue of A^k (k is a positive integer) is:

(A) k * lambda
(B) lambda^k
(C) lambda / k
(D) lambda + k

17. The eigenvalues of an upper triangular matrix are strictly its:

(A) Superdiagonal elements
(B) Main diagonal elements
(C) Subdiagonal elements
(D) First row entries

18. A system of linear equations AX = B is called consistent if it possesses:

(A) Exactly zero solutions
(B) At least one solution
(C) Only the trivial solution
(D) Infinitely many imaginary solutions

19. According to the Rouche-Capelli theorem, AX = B is consistent if and only if:

(A) rank(A) > rank([A|B])
(B) rank(A) = rank([A|B])
(C) rank(A) < rank([A|B])
(D) det(A) = det(B)

20. A consistent system AX = B with n unknowns has a unique solution if and only if:

(A) rank(A) = rank([A|B]) < n
(B) rank(A) = rank([A|B]) = n
(C) rank(A) != rank([A|B])
(D) det(A) = 0

21. A consistent system AX = B with n unknowns has infinitely many solutions if:

(A) rank(A) = rank([A|B]) = r < n
(B) rank(A) = n
(C) rank([A|B]) = n+1
(D) det(A) != 0

22. A homogeneous system of linear equations AX = 0 always possesses:

(A) No solution
(B) At least the trivial solution X = 0
(C) Only complex solutions
(D) Two distinct solutions

23. For a homogeneous system AX = 0 with n unknowns, non-trivial solutions exist if and only if:

(A) rank(A) = n
(B) rank(A) < n
(C) det(A) != 0
(D) A is identity

24. The Cayley-Hamilton theorem states that every square matrix satisfies its own:

(A) Eigenvector equation
(B) Characteristic equation
(C) Adjoint equation
(D) Diagonal form

25. If the characteristic equation of matrix A is lambda^2 - 5*lambda + 6 = 0, then A^-1 is:

(A) (1/6)*(5*I - A)
(B) (1/5)*(6*I - A)
(C) 5*A - 6*I
(D) (1/6)*(A - 5*I)

26. Eigenvectors corresponding to distinct eigenvalues of a real matrix are always:

(A) Linearly dependent
(B) Linearly independent
(C) Orthogonal
(D) Identical

27. A real symmetric matrix always possesses eigenvalues that are:

(A) Purely imaginary
(B) Real
(C) Complex conjugate pairs
(D) All zero

28. The eigenvectors of a real symmetric matrix corresponding to distinct eigenvalues are mutually:

(A) Collinear
(B) Orthogonal
(C) Parallel
(D) Zero vectors

29. A square matrix A is diagonalizable if and only if it has:
- **(A)** Zero trace
- **(B)** n linearly independent eigenvectors
- **(C)** All eigenvalues equal to 1
- **(D)** det(A) = 0

30. If P diagonalizes matrix A such that D = P^-1 * A * P, the columns of P are:
- **(A)** Rows of A
- **(B)** Eigenvectors of A
- **(C)** Pivots of A
- **(D)** Orthogonal bases of null space

31. The geometric multiplicity of an eigenvalue lambda is defined as the:
- **(A)** Multiplicity of lambda as a root of det(A - lambda*I) = 0
- **(B)** Dimension of the eigenspace Null(A - lambda*I)
- **(C)** Trace of matrix A
- **(D)** Rank of matrix A

32. The algebraic multiplicity (AM) and geometric multiplicity (GM) of an eigenvalue satisfy:
- **(A)** GM > AM
- **(B)** 1 <= GM <= AM
- **(C)** GM = AM + 1
- **(D)** GM * AM = 0

33. A quadratic form Q(x) = X^T * A * X is called positive definite if Q(x) > 0 for all:
- **(A)** x in R^n
- **(B)** x != 0 in R^n
- **(C)** x = 0 only
- **(D)** x with norm 1

34. By Sylvester's criterion, a symmetric matrix is positive definite if and only if:
- **(A)** All its elements are positive
- **(B)** All leading principal minors are strictly positive
- **(C)** Trace is positive
- **(D)** Determinant is negative

35. The signature of a real quadratic form is defined as the difference between the number of:
- **(A)** Rows and columns
- **(B)** Positive and negative eigenvalues
- **(C)** Pivots and free variables
- **(D)** Zero and non-zero rows

36. In biotechnology, a Leslie matrix is utilized to model:
- **(A)** Substrate enzyme kinetics
- **(B)** Age-structured population dynamics
- **(C)** DNA base pair sequences
- **(D)** Chromatography flow rate

37. In metabolic flux analysis, the stoichiometric matrix S satisfies steady-state balance:
- **(A)** S * v = 0
- **(B)** S + v = 1
- **(C)** S * v = infinity
- **(D)** det(S) = v

38. The rank-nullity theorem for an m x n matrix A states that rank(A) + nullity(A) equals:
- **(A)** m
- **(B)** n
- **(C)** m + n
- **(D)** m * n

39. If A is an idempotent matrix (A^2 = A), its eigenvalues can only be:
- **(A)** 1 and -1
- **(B)** 0 and 1
- **(C)** 0 and -1
- **(D)** Any positive integer

40. If A is an involuntary matrix (A^2 = I), its eigenvalues can only be:
- **(A)** 0 and 1
- **(B)** +1 and -1
- **(C)** 0 and -1
- **(D)** 2 and -2

41. The rank of the matrix A = [1 2 3; 2 4 6; 3 6 9] is:
- **(A)** 3
- **(B)** 2
- **(C)** 1
- **(D)** 0

42. The rank of the matrix A = [1 1 1; 1 2 3; 1 4 9] is:
- **(A)** 1
- **(B)** 2
- **(C)** 3
- **(D)** 0

43. For what value of k does the system x + y + z = 1, x + 2y + 4z = k, x + 4y + 10z = k^2 have a solution?

(A) k = 1 or k = 2
(B) k = 2 or k = 3
(C) Any real k
(D) No real k

44. If matrix A = [2 1; 1 2], the eigenvalues of A are:

(A) 1 and 2
(B) 1 and 3
(C) 2 and 3
(D) 0 and 3

45. For matrix A = [3 1; 0 3], the algebraic and geometric multiplicities of lambda = 3 are:

(A) AM = 1, GM = 1
(B) AM = 2, GM = 1
(C) AM = 2, GM = 2
(D) AM = 1, GM = 2

46. Is the matrix A = [3 1; 0 3] diagonalizable over R?

(A) Yes, always
(B) No, because GM < AM
(C) Yes, because det(A) != 0
(D) No, because trace is 6

47. The eigenvalues of matrix A = [1 2; 3 2] are:

(A) 4 and -1
(B) 3 and 2
(C) -4 and 1
(D) 5 and -1

48. The normalized eigenvector of A = [2 0; 0 3] corresponding to lambda = 2 is:

(A) [1, 0]^T
(B) [0, 1]^T
(C) [1, 1]^T / sqrt(2)
(D) [1, -1]^T

49. Using Cayley-Hamilton for A = [1 2; 0 2], the expression for A^3 in terms of A and I is:

(A) 7*A - 6*I
(B) 3*A - 2*I
(C) 6*A - 7*I
(D) 8*A - 6*I

50. If A is a 3x3 matrix with eigenvalues 1, -2, 3, then det(A^2 - 2*A) equals:

(A) 0
(B) -15
(C) 45
(D) 18

51. The quadratic form Q(x1, x2) = 2*x1^2 + 4*x1*x2 + 5*x2^2 has matrix representation:

(A) [2 4; 4 5]
(B) [2 2; 2 5]
(C) [4 2; 2 5]
(D) [2 0; 0 5]

52. The nature of the quadratic form Q(x, y) = 2*x^2 + 2*x*y + 2*y^2 is:

(A) Negative definite
(B) Positive definite
(C) Indefinite
(D) Semidefinite

53. The eigenvalues of the matrix A = [0 1; -1 0] are:

(A) 1 and -1
(B) i and -i
(C) 0 and 1
(D) 2 and -2

54. If A is a 3x3 real orthogonal matrix and det(A) = 1, then A represents a:

(A) Pure reflection
(B) Pure rotation in R^3
(C) Scaling transformation
(D) Shear transformation

55. The trace of matrix A = [5 2 -1; 3 0 4; 1 6 -2] is:

(A) 5
(B) 3
(C) 7
(D) -2

56. The system x + 2y - z = 0, 2x + y + z = 0, x - y + 2z = 0 has how many linearly independent solutions?

(A) 0
(B) 1
(C) 2
(D) 3

57. If lambda = 0 is an eigenvalue of matrix A, which of the following is strictly true?

(A) A is non-singular
(B) det(A) = 0
(C) A is symmetric
(D) Trace(A) = 0

58. If A and B are square matrices of the same order, then Trace(A * B) equals:

(A) Trace(A) * Trace(B)
(B) Trace(B * A)
(C) Trace(A) + Trace(B)
(D) det(A * B)

59. For an n x n matrix A, if A^k = 0 for some integer k >= 1 (nilpotent), all eigenvalues of A are:

(A) 1
(B) 0
(C) -1
(D) k

60. The modal matrix P that diagonalizes A = [1 1; 0 2] is composed of columns:

(A) [1, 0]^T and [1, 1]^T
(B) [1, 1]^T and [0, 1]^T
(C) [1, -1]^T and [1, 1]^T
(D) [2, 0]^T and [0, 1]^T

61. The rank of the augmented matrix [A|B] for an inconsistent system is always:

(A) Equal to rank(A)
(B) Greater than rank(A)
(C) Less than rank(A)
(D) Zero

62. In Gauss-Jordan elimination, the coefficient matrix is transformed into:

(A) Upper triangular form
(B) Lower triangular form
(C) Reduced row echelon form (identity)
(D) Tridiagonal form

63. The number of independent parameters in the general solution of AX = 0 with rank r and n variables is:

(A) r
(B) n
(C) n - r
(D) n + r

64. If A = [a b; c d], the characteristic equation det(A - lambda*I) = 0 can be written as:

(A) lambda^2 - Trace(A)*lambda + det(A) = 0
(B) lambda^2 + Trace(A)*lambda - det(A) = 0
(C) lambda^2 - det(A)*lambda + Trace(A) = 0
(D) lambda^2 - Trace(A) = 0

65. If the eigenvalues of A are 2, 4, 8, the eigenvalues of adj(A) are:

(A) 1/2, 1/4, 1/8
(B) 32, 16, 8
(C) 64, 32, 16
(D) 4, 8, 16

66. The quadratic form Q(x, y, z) = x^2 + 2*y^2 + 3*z^2 has rank and signature:

(A) Rank 3, Signature 3
(B) Rank 3, Signature 1
(C) Rank 2, Signature 2
(D) Rank 1, Signature 1

67. A biotechnology microbial growth system has state vector x_{k+1} = L * x_k. The long-term growth rate is determined by:

(A) The minimum eigenvalue of L
(B) The strictly dominant eigenvalue of L (spectral radius)
(C) Trace of L
(D) Determinant of L

68. In metabolic network stoichiometry S * v = 0, the degrees of freedom for flux variability is given by:

(A) dim(Col(S))
(B) Nullity of S (dim(Null(S)))
(C) Rank of S
(D) Trace of S

69. If A is symmetric, any two eigenvectors corresponding to distinct eigenvalues lambda1 != lambda2 satisfy:

(A) X1 + X2 = 0
(B) X1^T * X2 = 0
(C) X1 = c * X2
(D) X1^T * X2 = 1

70. If A is skew-symmetric of odd order 2k+1, its determinant is always:

(A) 1
(B) 0
(C) -1
(D) Odd integer

71. The inverse of an orthogonal matrix Q is given by:
- **(A)** Q^-1 = -Q
- **(B)** Q^-1 = Q^T
- **(C)** Q^-1 = det(Q)*Q
- **(D)** Q^-1 = adj(Q)

72. The spectral radius rho(A) of a square matrix A is defined as:
- **(A)** min |lambda_i|
- **(B)** max |lambda_i|
- **(C)** Sum of |lambda_i|
- **(D)** Product of |lambda_i|

73. For the system 2x + y = 5, 4x + 2y = 10, the system is:
- **(A)** Inconsistent
- **(B)** Consistent with unique solution
- **(C)** Consistent with infinitely many solutions
- **(D)** Trivial

74. For the system x + y = 2, 2x + 2y = 5, the system is:
- **(A)** Consistent with unique solution
- **(B)** Inconsistent
- **(C)** Consistent with infinite solutions
- **(D)** Indeterminate

75. The product of two symmetric matrices A and B is symmetric if and only if:
- **(A)** A and B commute (A * B = B * A)
- **(B)** det(A) = det(B)
- **(C)** A = B^-1
- **(D)** Trace(A) = Trace(B)

76. Evaluate the rank of the 4x4 matrix A where a_ij = i * j:
- **(A)** 4
- **(B)** 2
- **(C)** 1
- **(D)** 0

77. Find the value of 'a' for which matrix A = [1 1 1; 1 2 4; 1 3 a] has rank 2:
- **(A)** 5
- **(B)** 6
- **(C)** 7
- **(D)** 9

78. Let A be a 3x3 matrix with eigenvalues 1, 2, 3. What is the trace of A^3?
- **(A)** 14
- **(B)** 36
- **(C)** 24
- **(D)** 32

79. If A = [1 2; 3 4], using Cayley-Hamilton theorem, find A^2 - 5*A - 2*I:
- **(A)** I
- **(B)** O (Zero matrix)
- **(C)** 2*I
- **(D)** 5*A

80. Given A = [2 2; 1 3], what is the matrix A^4?
- **(A)** [256 254; 127 383]
- **(B)** [170 170; 85 255]
- **(C)** [128 128; 64 192]
- **(D)** [300 200; 100 400]

81. The eigenvalues of the 3x3 matrix A = [2 1 1; 1 2 1; 1 1 2] are:
- **(A)** 1, 1, 4
- **(B)** 2, 2, 2
- **(C)** 0, 1, 5
- **(D)** 1, 2, 3

82. For matrix A = [2 1 1; 1 2 1; 1 1 2], what is the dimension of the eigenspace corresponding to lambda = 1?
- **(A)** 1
- **(B)** 2
- **(C)** 3
- **(D)** 0

83. What is the determinant of the 4x4 circulant matrix whose rows are cyclic shifts of [1, 2, 3, 4]?
- **(A)** -160
- **(B)** 160
- **(C)** 0
- **(D)** 256

84. If A is a 3x3 matrix with det(A) = 4, then det(adj(adj(A))) equals:
- **(A)** 16
- **(B)** 64
- **(C)** 256
- **(D)** 4

85. Find the quadratic form associated with symmetric matrix A = [1 2 3; 2 0 4; 3 4 5]:
(A) x^2 + 5z^2 + 4xy + 6xz + 8yz
(B) x^2 + 5z^2 + 2xy + 3xz + 4yz
(C) x^2 + 4xy + 5z^2
(D) 2x^2 + 3y^2 + 5z^2 + 8yz

86. The quadratic form Q = x1^2 + 2*x2^2 + 3*x3^2 + 2*x1*x2 - 2*x1*x3 + 2*x2*x3 has principal minors Delta1, Delta2, Delta3 equal to:
(A) 1, 1, -2 (Indefinite)
(B) 1, 1, 0 (Positive semidefinite)
(C) 1, 2, 6 (Positive definite)
(D) 1, -1, 3 (Indefinite)

87. In a 2-stage age structured population Leslie matrix L = [0 4; 0.5 0], the dominant eigenvalue is:
(A) 1.0
(B) sqrt(2) = 1.414
(C) 2.0
(D) 4.0

88. For the Leslie matrix L = [0 4; 0.5 0], the stable stage ratio (eigenvector for lambda = sqrt(2)) is:
(A) [2*sqrt(2), 1]^T
(B) [4, 1]^T
(C) [1, 2]^T
(D) [sqrt(2), 1]^T

89. If A = [1 4; 2 3], the inverse A^-1 computed via Cayley-Hamilton theorem is:
(A) (-1/5) * [3 -4; -2 1]
(B) (1/5) * [3 -4; -2 1]
(C) (-1/5) * [-3 4; 2 -1]
(D) (1/5) * [1 4; 2 3]

90. If A is an n x n skew-symmetric matrix and n is even, then det(A) is always:
(A) 0
(B) A perfect square
(C) Negative
(D) Purely imaginary

91. The linear transformation T: R^3 -> R^3 has matrix [1 2 3; 0 1 4; 5 6 0]. The dimension of the Range of T is:
(A) 1
(B) 2
(C) 3
(D) 0

92. If A is an invertible matrix and lambda is an eigenvalue of A with eigenvector X, then X is an eigenvector of (A + 3*I)^-1 with eigenvalue:
(A) 1 / (lambda + 3)
(B) lambda + 3
(C) 3 / lambda
(D) lambda / (lambda + 3)

93. The condition number kappa(A) = ||A|| * ||A^-1|| of an orthogonal matrix is:
(A) 0
(B) 1
(C) Infinity
(D) 2

94. Solve the system using Cramer's rule: 2x - y = 1, x + 3y = 11. The value of x is:
(A) 1
(B) 2
(C) 3
(D) 4

95. Solve the system 2x - y = 1, x + 3y = 11. The value of y is:
(A) 1
(B) 2
(C) 3
(D) 4

96. For what value of lambda does the system (lambda - 1)x + y = 0, x + (lambda - 1)y = 0 have non-trivial solutions?
(A) lambda = 0, 2
(B) lambda = 1, -1
(C) lambda = 2, 3
(D) lambda = 0, 1

97. If A = [3 -2; 1 0], what is the expression for A^n using diagonalization?
(A) (2^n - 1)*A - (2^n - 2)*I
(B) 2^n * A - I
(C) (2^n - 1)*I + A
(D) A + n*I

98. A metabolic network contains 4 metabolites and 6 reaction fluxes. The maximum number of independent steady-state flux pathways is:
(A) 2
(B) 4
(C) 6
(D) 10

99. The matrix A = [1 0 0; 0 cos(theta) -sin(theta); 0 sin(theta) cos(theta)] has eigenvalues:

(A) 1, e^(i*theta), e^(-i*theta)
(B) 1, 1, 1
(C) 0, cos(theta), sin(theta)
(D) 1, -1, 1

100. If A is a positive definite symmetric matrix, then all diagonal entries a_ii must be:

(A) Negative
(B) Strictly positive (a_ii > 0)
(C) Zero
(D) Complex numbers

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Define elementary row operations on matrices, explain Row Echelon Form (REF) and Reduced Row Echelon Form (RREF), and prove that elementary operations preserve matrix rank.

Q2. Determine the rank of the following 4x4 matrix A by systematically reducing it to Row Echelon Form: A = [[1, 2, -1, 3], [2, 4, 1, -2], [3, 6, 3, -7], [1, 2, 2, -5]].

Q3. State the Rouche-Capelli Theorem governing the consistency of a non-homogeneous system AX = B, and provide a comprehensive theoretical proof.

Q4. Investigate the consistency of the following system of equations for different values of lambda and mu: x + y + z = 6; x + 2y + 3z = 10; x + 2y + lambda*z = mu. Find the conditions for: (i) No solution, (ii) A unique solution, (iii) Infinitely many solutions.

Q5. Explain the structure of the solution space of a homogeneous system of linear equations AX = 0. Define nullity and prove the Rank-Nullity Theorem for an m x n matrix A.

Q6. Describe the Gauss-Jordan Elimination method for matrix inversion. Using this method, compute the inverse of matrix A = [[1, 2, 3], [2, 5, 3], [1, 0, 8]].

Q7. Solve the following system of 4 linear equations using Gaussian elimination with back substitution: 2x1 + x2 + 3x3 - x4 = 4; x1 + 2x2 - x3 + 2x4 = 5; 3x1 - x2 + 2x3 + x4 = 7; 4x1 + 3x2 - x3 - 2x4 = 1.

Q8. Explain the Gauss-Seidel iterative method for solving AX = B. State the sufficient condition for convergence (diagonally dominant matrix) and perform 3 iterations for the system: 8x - 3y + 2z = 20; 4x + 11y - z = 33; 6x + 3y + 12z = 35, starting from (0, 0, 0).

Q9. Define Eigenvalues and Eigenvectors of a linear operator. Prove that eigenvectors corresponding to distinct eigenvalues of any square matrix are linearly independent.

Q10. Prove the three fundamental spectral properties of eigenvalues: (i) Sum of eigenvalues equals the Trace of the matrix, (ii) Product of eigenvalues equals the Determinant, (iii) If lambda is an eigenvalue of A, then 1/lambda is an eigenvalue of A^-1.

Q11. Prove that for any real symmetric matrix A: (i) All its eigenvalues are strictly real, and (ii) Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.

Q12. Find all eigenvalues and corresponding normalized eigenvectors of the real symmetric matrix: A = [[3, -1, 1], [-1, 5, -1], [1, -1, 3]]. Verify that the eigenvectors are mutually orthogonal.

Q13. State the Cayley-Hamilton Theorem. Provide a rigorous algebraic proof for an arbitrary 2x2 matrix, and explain its general proof for an n x n matrix using the adjoint matrix identity.

Q14. Verify the Cayley-Hamilton Theorem for matrix A = [[1, 0, 3], [2, 1, -1], [1, -1, 1]], and hence compute A^-1 and A^4.

Q15. Explain the criteria for diagonalizability of a square matrix. Define modal matrix P and spectral matrix D, and show that A^k = P * D^k * P^-1.

Q16. Diagonalize the matrix A = [[1, 6, 1], [0, 2, 0], [0, 0, 3]] and compute A^5 using matrix diagonalization.

Q17. Define an orthogonal matrix and an orthogonal transformation. Prove that orthogonal transformations preserve: (i) Vector lengths (Euclidean norm), and (ii) Inner products (angles between vectors).

Q18. State and prove the Spectral Decomposition Theorem for a real symmetric matrix A. Show how A can be written as a sum of rank-one projection matrices.

Q19. Define a quadratic form, its matrix representation, rank, index, signature, and classifications (positive definite, negative definite, indefinite, semidefinite).

Q20. Reduce the quadratic form Q = 3*x1^2 + 5*x2^2 + 3*x3^2 - 2*x1*x2 + 2*x1*x3 - 2*x2*x3 to canonical form by orthogonal transformation. Determine its rank, index, signature, and nature.

Q21. State and prove Sylvester's Criterion for positive definiteness of a real symmetric matrix.

Q22. Define a linear transformation T: V -> W. Define its Kernel and Range. Find the matrix representation of T: P2(R) -> P2(R) defined by T(p(x)) = p'(x) + p''(x) with respect to standard basis {1, x, x^2}.

Q23. State and prove the Rank-Nullity Theorem (Dimension Theorem) for an arbitrary linear transformation T: V -> W between finite-dimensional vector spaces.

Q24. Derive Cramer's Rule for solving a system of n linear equations in n unknowns. Solve the following 3x3 system using Cramer's rule: 2x - y + 3z = 9; x + y + z = 6; x - y + z = 2.

Q25. Explain LU Decomposition (Doolittle factorization) of a square matrix. Factorize matrix A = [[2, 1, 1], [4, 1, 0], [-2, 2, 1]] into L and U, and solve AX = [1, -2, 7]^T.

Q26. Explain the mathematical theory of Singular Value Decomposition (SVD). Define singular values and express the pseudo-inverse (Moore-Penrose inverse) A^+ in terms of SVD.

Q27. Describe the Power Method algorithm for computing the dominant eigenvalue and eigenvector of a matrix. Apply the method for 4 iterations to matrix A = [[4, 1], [1, 3]] starting with initial vector X0 = [1, 0]^T.

Q28. Formulate the Leslie Matrix population model in biotechnology for age-structured microbial or animal cell cultures. Derive the characteristic equation for the intrinsic growth rate and explain the stable age distribution.

Q29. In Metabolic Flux Analysis (MFA), explain how stoichiometric matrices and null space analysis are used to determine steady-state metabolic reaction rates. Illustrate with a simple 3-metabolite, 4-flux metabolic pathway.

Q30. Model a multi-component bioprocess separation column using a linear system of differential equations dX/dt = A*X. Solve the system using the matrix exponential e^(At) where A = [[-2, 1], [0, -3]] and X(0) = [100, 50]^T.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[C]	**Q61:**	[B]	**Q81:**	[A]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[C]	**Q62:**	[C]	**Q82:**	[B]
Q3:	[B]	**Q23:**	[B]	**Q43:**	[A]	**Q63:**	[C]	**Q83:**	[A]
Q4:	[B]	**Q24:**	[B]	**Q44:**	[B]	**Q64:**	[A]	**Q84:**	[C]
Q5:	[B]	**Q25:**	[A]	**Q45:**	[B]	**Q65:**	[B]	**Q85:**	[A]
Q6:	[B]	**Q26:**	[B]	**Q46:**	[B]	**Q66:**	[A]	**Q86:**	[B]
Q7:	[C]	**Q27:**	[B]	**Q47:**	[A]	**Q67:**	[B]	**Q87:**	[B]
Q8:	[C]	**Q28:**	[B]	**Q48:**	[A]	**Q68:**	[B]	**Q88:**	[A]
Q9:	[B]	**Q29:**	[B]	**Q49:**	[A]	**Q69:**	[B]	**Q89:**	[A]
Q10:	[B]	**Q30:**	[B]	**Q50:**	[A]	**Q70:**	[B]	**Q90:**	[B]
Q11:	[C]	**Q31:**	[B]	**Q51:**	[B]	**Q71:**	[B]	**Q91:**	[C]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[B]	**Q72:**	[B]	**Q92:**	[A]
Q13:	[C]	**Q33:**	[B]	**Q53:**	[B]	**Q73:**	[C]	**Q93:**	[B]
Q14:	[B]	**Q34:**	[B]	**Q54:**	[B]	**Q74:**	[B]	**Q94:**	[B]
Q15:	[B]	**Q35:**	[B]	**Q55:**	[B]	**Q75:**	[A]	**Q95:**	[C]
Q16:	[B]	**Q36:**	[B]	**Q56:**	[B]	**Q76:**	[C]	**Q96:**	[A]
Q17:	[B]	**Q37:**	[A]	**Q57:**	[B]	**Q77:**	[C]	**Q97:**	[A]
Q18:	[B]	**Q38:**	[B]	**Q58:**	[B]	**Q78:**	[B]	**Q98:**	[A]
Q19:	[B]	**Q39:**	[B]	**Q59:**	[B]	**Q79:**	[B]	**Q99:**	[A]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[B]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[0 0 6 -16]
[0 0 3 -8]
Step 3: In row 2, the pivot is a23 = 3. Eliminate entries in column 3 below row 2:
Operation R3 -> R3 - 2*R2:
[0, 0, 6 - 2(3), -16 - 2(-8)] = [0, 0, 0, 0]
Operation R4 -> R4 - 1*R2:
[0, 0, 3 - 1(3), -8 - 1(-8)] = [0, 0, 0, 0]
Matrix after Step 3:
[1 2 -1 3]
REF = [0 0 3 -8]
[0 0 0 0]
[0 0 0 0]
Step 4: Count non-zero rows:
Row 1 is non-zero (pivot at column 1).
Row 2 is non-zero (pivot at column 3).
Rows 3 and 4 are entirely zero.
The number of non-zero rows in the Row Echelon Form is 2.
Conclusion:
Rank(A) = rho(A) = 2.

>>**Solution to Question 3:** State the Rouche-Capelli Theorem governing the consistency of a non-homogeneous system AX = B, and provide a comprehensive th

- For unique vs infinite solutions: By the Rank-Nullity theorem, dim(Null(A)) = n - rank(A) = n - r.

If r = n, Null(A) = {0}, giving a unique particular solution X = X_p.

If r < n, general solution is X = X_p + X_h, where X_h spans an (n - r)-dimensional subspace, yielding infinitely many solutions. Hence proved.

[Detailed Step-by-Step Answer]:
1. Structure of Solution Space of AX = 0:
- Let A in M_{m x n}(R). The set of all solutions to AX = 0 is defined as the Null Space (Kernel) of A:
Null(A) = { X in R^n : AX = 0 }.
- Null(A) is a subspace of R^n because:
(a) Zero vector: A * 0 = 0 => 0 in Null(A).
(b) Additive closure: If X1, X2 in Null(A), then A(X1 + X2) = AX1 + AX2 = 0 + 0 = 0 => X1 + X2 in Null(A).
(c) Scalar closure: If X in Null(A) and c in R, then A(c*X) = c*(AX) = c*0 = 0 => c*X in Null(A).
- Nullity of A is defined as the dimension of Null(A): nullity(A) = dim(Null(A)).
2. Formal Statement of Rank-Nullity Theorem:
For any m x n matrix A:
rank(A) + nullity(A) = n (the total number of columns / variables).
3. Mathematical Proof:
- Reduce A to Reduced Row Echelon Form (RREF) R via elementary row operations.
- Let rank(A) = r. Then R has exactly r non-zero rows, each containing a leading 1 (pivot).
- The n variables x1, x2, ..., xn are partitioned into two disjoint subsets:
* r Pivot (Basic) variables: corresponding to columns containing leading 1s.
* (n - r) Free variables: corresponding to columns without leading 1s. Let these be t1, t2, ..., t_{n-r}.
- Each basic variable can be uniquely expressed as a linear combination of the (n - r) free variables.
- Setting one free variable to 1 and all others to 0 generates a solution vector V_j (for j = 1, 2, ..., n - r).
- These (n - r) vectors {V1, V2, ..., V_{n-r}} are linearly independent because in the free variable positions, they contain the columns of the (n - r) x (n - r) identity matrix.
- Any solution X to AX = 0 is a linear combination of {V1, ..., V_{n-r}}:
X = t1*V1 + t2*V2 + ... + t_{n-r}*V_{n-r}.
- Therefore, {V1, ..., V_{n-r}} forms a basis for Null(A).
- Consequently, dim(Null(A)) = nullity(A) = n - r.
- Since r = rank(A), we obtain: rank(A) + nullity(A) = r + (n - r) = n. Hence proved.

>> Solution to Question 6: Describe the Gauss-Jordan Elimination method for matrix inversion. Using this method, compute the inverse of matrix A = [[1, 2, 3], [2,

Matrix becomes:
[1 2 3 | 1 0 0]
[0 1 -3 | -2 1 0]
[0 -2 5 | -1 0 1]
Step 3: Eliminate in column 2 using pivot a22 = 1:
R1 -> R1 - 2*R2: [1 0 9 | 5 -2 0]
R3 -> R3 + 2*R2: [0 0 -1 | -5 2 1]
Matrix becomes:
[1 0 9 | 5 -2 0]
[0 1 -3 | -2 1 0]
[0 0 -1 | -5 2 1]
Step 4: Make pivot a33 = 1 and eliminate in column 3:
R3 -> (-1)*R3: [0 0 1 | 5 -2 -1]
Now eliminate above pivot a33:
R1 -> R1 - 9*R3:
[1, 0, 9 - 9(1) | 5 - 9(5), -2 - 9(-2), 0 - 9(-1)] = [1 0 0 | -40 16 9]
R2 -> R2 + 3*R3:
[0, 1, -3 + 3(1) | -2 + 3(5), 1 + 3(-2), 0 + 3(-1)] = [0 1 0 | 13 -5 -3]
Step 5: Read out inverse matrix A^-1:
A^-1 = [-40 16 9]
[13 -5 -3]
[5 -2 -1]
Verification:
A * A^-1:
Row 1 * Col 1: 1(-40) + 2(13) + 3(5) = -40 + 26 + 15 = 1.
Row 1 * Col 2: 1(16) + 2(-5) + 3(-2) = 16 - 10 - 6 = 0.
Row 2 * Col 2: 2(16) + 5(-5) + 3(-2) = 32 - 25 - 6 = 1. (Confirmed correct).

>>Solution to Question 67: Solve the following system of 4 linear equations using Gaussian elimination with back substitution: 2x1 + x2 + 3x3 - x4 = 4; x1 + 2x2 - x

3*[0, -5, 3, -10 | -19] - 5*[0, -3, 5, -5 | 6]
= [0, -15+15, 9-25, -30+25 | -57-30] = [0 0 -16 -5 | -87]
Step 5: Eliminate column 3 in row 4:
From R3: 10*x3 - 10*x4 = 33 => x3 - x4 = 3.3 => x3 = x4 + 3.3.
Substitute into R4: -16(x4 + 3.3) - 5*x4 = -87
=> -21*x4 - 52.8 = -87 => -21*x4 = -34.2 => x4 = 34.2 / 21 = 1.6286.
Step 6: Back-substitution:
x4 = 1.6286
x3 = 1.6286 + 3.3 = 4.9286
From R2: -3*x2 + 5(4.9286) - 5(1.6286) = -6 => -3*x2 + 16.5 = -6 => -3*x2 = -22.5 => x2 = 7.5
From R1: x1 + 2(7.5) - 4.9286 + 2(1.6286) = 5
=> x1 + 15 - 4.9286 + 3.2572 = 5 => x1 + 13.3286 = 5 => x1 = -8.3286.
Final Solution:
x1 = -8.329, x2 = 7.500, x3 = 4.929, x4 = 1.629.

>>Solution to Question 8: Explain the Gauss-Seidel iterative method for solving AX = B. State the sufficient condition for convergence (diagonally dominant matrix).

After 3 iterations, the solution converges rapidly towards: x ~ 3.00, y ~ 2.00, z ~ 0.91.

>>Solution Question 46 Eigenvalues and Eigenvectors of a linear operator: Prove that eigenvectors corresponding to distinct eigenvalues of a matrix are linearly

- Let A be an n x n matrix. The characteristic polynomial is defined as:
p(lambda) = det(lambda * I - A) = lambda^n - c_{n-1}*lambda^{n-1} + ... + (-1)^n * c0.
- By definition of determinant, the only term in det(lambda*I - A) containing lambda^n and lambda^{n-1} comes from the product of the diagonal elements:
Product_{i=1}^n (lambda - a_ii) = lambda^n - (sum_{i=1}^n a_ii) * lambda^{n-1} + ...
- Therefore, the coefficient of lambda^{n-1} is: -sum_{i=1}^n a_ii = -Trace(A).
- By the fundamental theorem of algebra, p(lambda) factors completely over C into its roots (eigenvalues):
p(lambda) = (lambda - lambda_1) * (lambda - lambda_2) * ... * (lambda - lambda_n)
= lambda^n - (sum_{i=1}^n lambda_i) * lambda^{n-1} + ... + (-1)^n * Product_{i=1}^n lambda_i.
- Equating coefficients of lambda^{n-1}:
-(sum_{i=1}^n lambda_i) = -Trace(A) => sum_{i=1}^n lambda_i = Trace(A). (Proved).
2. Proof that Product of Eigenvalues = det(A):
- Set lambda = 0 in both expressions of the characteristic polynomial:
From factorization: p(0) = (0 - lambda_1)(0 - lambda_2)...(0 - lambda_n) = (-1)^n * Product_{i=1}^n lambda_i.
From determinant definition: p(0) = det(0*I - A) = det(-A) = (-1)^n * det(A).
- Equating both expressions:
(-1)^n * Product_{i=1}^n lambda_i = (-1)^n * det(A) => Product_{i=1}^n lambda_i = det(A). (Proved).
3. Proof for Eigenvalues of A^-1:
- Let A be invertible (det(A) != 0, so lambda != 0).
- Let X != 0 be an eigenvector corresponding to eigenvalue lambda: A * X = lambda * X.
- Multiply both sides from the left by A^-1:
A^-1 * (A * X) = A^-1 * (lambda * X)
=> I * X = lambda * (A^-1 * X)
=> X = lambda * (A^-1 * X).
- Since lambda != 0, divide both sides by scalar lambda:
A^-1 * X = (1 / lambda) * X.
- This matches the eigenvector definition: X is an eigenvector of A^-1 with eigenvalue 1/lambda. (Proved).

>>Solution Statement: Prove that for any real symmetric matrix A: (i) All its eigenvalues are strictly real, and (ii) Eigenvectors corresponding to distinct eigenva

- A complex number equals its conjugate if and only if its imaginary part is zero.
- Hence, lambda is strictly REAL. (Proved).

2. Proof of Orthogonality of Eigenvectors:
- Let lambda_1 and lambda_2 be two distinct eigenvalues (lambda_1 != lambda_2) of real symmetric matrix A.
- Let X_1 and X_2 be corresponding real eigenvectors:

A * X_1 = lambda_1 * X_1 and A * X_2 = lambda_2 * X_2.
- Consider the scalar quantity X_1^T * A * X_2:

X_1^T * (A * X_2) = X_1^T * (lambda_2 * X_2) = lambda_2 * (X_1^T * X_2).
- Also, using the transpose property (A * X_1)^T = X_1^T * A^T = X_1^T * A:

(X_1^T * A) * X_2 = (A * X_1)^T * X_2 = (lambda_1 * X_1)^T * X_2 = lambda_1 * (X_1^T * X_2).
- Subtracting the two expressions:

lambda_1 * (X_1^T * X_2) - lambda_2 * (X_1^T * X_2) = 0

=> (lambda_1 - lambda_2) * (X_1^T * X_2) = 0.
- Since lambda_1 != lambda_2, we have (lambda_1 - lambda_2) != 0.
- Consequently: X_1^T * X_2 = 0.
- This signifies that the Euclidean inner product <X1, X2> = 0.
- Hence, eigenvectors X_1 and X_2 are mutually ORTHOGONAL. (Proved).

>> Solution to Question 12: Find all eigenvalues and corresponding normalized eigenvectors of the real symmetric matrix: A = [[3, -1, 1], [-1, 5, -1], [1, -1, 3]]. Verify

x1 / (1 - (-1)) = x2 / (-1 - 3) = x3 / (3 - 1)
=> x1 / 2 = x2 / (-4) = x3 / 2 => x1 / 1 = x2 / (-2) = x3 / 1.
X3 = [1, -2, 1]^T. Norm ||X3|| = sqrt(1 + 4 + 1) = sqrt(6).
Normalized u3 = [1/sqrt(6), -2/sqrt(6), 1/sqrt(6)]^T.
Step 5: Verification of Orthogonality:
u1 . u2 = (-1)(1) + (0)(1) + (1)(1) = -1 + 0 + 1 = 0.
u1 . u3 = (-1)(1) + (0)(-2) + (1)(1) = -1 + 0 + 1 = 0.
u2 . u3 = (1)(1) + (1)(-2) + (1)(1) = 1 - 2 + 1 = 0.
All pairs are mutually orthogonal.

Solution to Question 12: the Cayley-Hamilton Theorem. Provide a rigorous algebraic proof for an arbitrary 2x2 matrix, and explain its general proof for an

A^n * B_{n-1} = A^n
A^{n-1} * B_{n-2} - A^n * B_{n-1} = c_{n-1} * A^{n-1}
...
A * B0 - A^2 * B1 = c1 * A
-A * B0 = c0 * I.
- Summing all equations causes the left-hand side to telescope completely to O (Zero matrix):
O = A^n + c_{n-1}*A^{n-1} + ... + c1*A + c0*I = p(A). Hence proved in full generality.

>>Solution to Question 14 Verify the Cayley-Hamilton Theorem for matrix A = [[1, 0, 3], [2, 1, -1], [1, -1, 1]], and hence compute A^-1 and A^4.

From A^3 = 3*A^2 + A - 9*I:
Multiply by A: A^4 = 3*A^3 + A^2 - 9*A
Substitute A^3: A^4 = 3(3*A^2 + A - 9*I) + A^2 - 9*A = 10*A^2 - 6*A - 27*I
10*A^2 = [40 -30 60; 30 20 40; 0 -20 50]
-6*A = [-6 0 -18; -12 -6 6; -6 6 -6]
-27*I = [-27 0 0; 0 -27 0; 0 0 -27]
A^4 = [40-6-27, -30+0+0, 60-18+0; 30-12+0, 20-6-27, 40+6+0; 0-6+0, -20+6+0, 50-6-27]
= [7 -30 42]
[18 -13 46]
[-6 -14 17].

Solution to Question 16: Explain the criteria for diagonalizability of a square matrix. Define modal matrix P and spectral matrix D, and show that A^k = P * D^k * P^-1 for any positive integer k.

[Problem Statement]: Diagonalize the matrix A = [[1, 6, 1], [0, 2, 0], [0, 0, 3]] and compute A^5 using matrix diagonalization.
[Detailed Step-by-Step Answer]:
Step 1: Eigenvalues of A:
Since A is upper triangular, its eigenvalues are simply the diagonal entries:
lambda1 = 1, lambda2 = 2, lambda3 = 3.
Since all three eigenvalues are distinct, matrix A is guaranteed to be diagonalizable.
Step 2: Find Eigenvectors:
- For lambda1 = 1:
(A - 1*I)X = 0 => [0 6 1; 0 1 0; 0 0 2][x1; x2; x3] = [0; 0; 0].
Row 3: 2*x3 = 0 => x3 = 0.
Row 2: 1*x2 = 0 => x2 = 0.
Row 1: 0*x1 + 6(0) + 1(0) = 0 (x1 is free).
Let x1 = 1 => X1 = [1, 0, 0]^T.
- For lambda2 = 2:
(A - 2*I)X = 0 => [-1 6 1; 0 0 0; 0 0 1][x1; x2; x3] = [0; 0; 0].
Row 3: x3 = 0.
Row 1: -x1 + 6*x2 + 0 = 0 => x1 = 6*x2.
Let x2 = 1 => x1 = 6.
X2 = [6, 1, 0]^T.
- For lambda3 = 3:
(A - 3*I)X = 0 => [-2 6 1; 0 -1 0; 0 0 0][x1; x2; x3] = [0; 0; 0].
Row 2: -x2 = 0 => x2 = 0.
Row 1: -2*x1 + 6(0) + x3 = 0 => x3 = 2*x1.
Let x1 = 1 => x3 = 2.
X3 = [1, 0, 2]^T.
Step 3: Construct Modal Matrix P and Inverse P^-1:
P = [1 6 1]
[0 1 0]
[0 0 2]
det(P) = 1 * (1*2 - 0) = 2.
Adjoint of P:
Cofactors:
C11 = 2, C12 = 0, C13 = 0
C21 = -12, C22 = 2, C23 = 0
C31 = -1, C32 = 0, C33 = 1
P^-1 = (1/2) * [2 -12 -1; 0 2 0; 0 0 1] = [1 -6 -0.5; 0 1 0; 0 0 0.5].
Step 4: Spectral Matrix D and D^5:
D = diag(1, 2, 3)
D^5 = diag(1^5, 2^5, 3^5) = diag(1, 32, 243).
Step 5: Compute A^5 = P * D^5 * P^-1:
P * D^5 = [1 6 1; 0 1 0; 0 0 2] * [1 0 0; 0 32 0; 0 0 243] = [1, 192, 243; 0, 32, 0; 0, 0,
486].
Now multiply by P^-1:
A^5 = [1, 192, 243; 0, 32, 0; 0, 0, 486] * [1 -6 -0.5; 0 1 0; 0 0 0.5]
Row 1: [1*1 + 0 + 0, 1(-6) + 192(1) + 0, 1(-0.5) + 0 + 243(0.5)] = [1, 186, 121]
Row 2: [0, 32(1), 0] = [0, 32, 0]
Row 3: [0, 0, 486(0.5)] = [0, 0, 243]
Final Result:
A^5 = [1 186 121]
[0 32 0]
[0 0 243].

- Expand the matrix products:
Q * D = [u_1 | u_2 | ... | u_n] * diag(lambda_1, ..., lambda_n) = [lambda_1*u_1 | lambda_2*u_2 | ... | lambda_n*u_n].
Post-multiplying by Q^T = [u_1^T; u_2^T; ...; u_n^T]:
A = [lambda_1*u_1 | ... | lambda_n*u_n] * [u_1^T; ...; u_n^T] = sum_{i=1}^n lambda_i * (u_i * u_i^T).
3. Verification of Projection Properties:
- P_i^2 = (u_i * u_i^T) * (u_i * u_i^T) = u_i * (u_i^T * u_i) * u_i^T = u_i * (1) * u_i^T = P_i.
- For i != j: P_i * P_j = (u_i * u_i^T) * (u_j * u_j^T) = u_i * (u_i^T * u_j) * u_j^T = u_i * (0) * u_j^T = O.
- Resolution of identity: Q * Q^T = I => sum_{i=1}^n u_i * u_i^T = I.
- Rank: Since u_i != 0, dim(Col(u_i * u_i^T)) = 1, so each P_i has rank 1. Hence proved.

>> Solution Store Question Bank

A = [-1 5 -1]
[1 -1 3]
Step 2: Find the Eigenvalues of A:
det(A - lambda*I) = lambda^3 - S1*lambda^2 + S2*lambda - S3 = 0.
S1 = Trace(A) = 3 + 5 + 3 = 11.
S2 = Sum of principal minors = (15 - 1) + (9 - 1) + (15 - 1) = 14 + 8 + 14 = 36.
S3 = det(A) = 3(14) - (-1)(-3 - (-1)) + 1(1 - 5) = 42 - 2 - 4 = 36.
Equation: lambda^3 - 11*lambda^2 + 36*lambda - 36 = 0.
Factoring: (lambda - 2)(lambda - 3)(lambda - 6) = 0.
Eigenvalues: lambda1 = 2, lambda2 = 3, lambda3 = 6.
Step 3: Construct Normalized Eigenvectors:
- For lambda1 = 2: u1 = [-1/sqrt(2), 0, 1/sqrt(2)]^T.
- For lambda2 = 3: u2 = [1/sqrt(3), 1/sqrt(3), 1/sqrt(3)]^T.
- For lambda3 = 6: u3 = [1/sqrt(6), -2/sqrt(6), 1/sqrt(6)]^T.
Step 4: Orthogonal Transformation Matrix Q:
Q = [-1/sqrt(2) 1/sqrt(3) 1/sqrt(6)]
[0 1/sqrt(3) -2/sqrt(6)]
[1/sqrt(2) 1/sqrt(3) 1/sqrt(6)]
Transformation: X = Q * Y, where Y = [y1, y2, y3]^T.
Step 5: Canonical Form:
Q(Y) = Y^T * (Q^T * A * Q) * Y = Y^T * D * Y
Q(y1, y2, y3) = 2*y1^2 + 3*y2^2 + 6*y3^2.
Step 6: Properties and Classification:
- Rank (r) = 3 (number of non-zero terms).
- Index (p) = 3 (number of positive terms).
- Signature (s) = 2p - r = 2(3) - 3 = 3.
- Nature: Since all eigenvalues (2, 3, 6) are strictly positive, the quadratic form is strictly POSITIVE DEFINITE.

>> Solution to Question 21: and prove Sylvester's Criterion for positive definiteness of a real symmetric matrix.

> 0. True.
* Inductive Hypothesis: Assume true for matrices of order n - 1.
Let A be an n x n matrix with Delta_k > 0 for k = 1, ..., n.
By hypothesis, A_{n-1} is positive definite and invertible.
Partition A as:
A = [A_{n-1} b]
[b^T c], where b in R^{n-1}, c in R.
Perform block Gaussian elimination: Let S = c - b^T * (A_{n-1})^-1 * b (the Schur complement of A_{n-1}).
Determinant identity: det(A) = det(A_{n-1}) * S => Delta_n = Delta_{n-1} * S.
Since Delta_n > 0 and Delta_{n-1} > 0, we must have S > 0.
For any non-zero x = [u; v] in R^n:
x^T * A * x = (u + A_{n-1}^-1 * b * v)^T * A_{n-1} * (u + A_{n-1}^-1 * b * v) + S * v^2.
If v != 0, then S * v^2 > 0 and the first term is >= 0 (since A_{n-1} is positive definite), so x^T * A * x > 0.
If v = 0, then u != 0, so x^T * A * x = u^T * A_{n-1} * u > 0.
Therefore, x^T * A * x > 0 for all x != 0.
Hence, A is strictly positive definite. (Proved).

>>Solution to Question 22: Let T: V -> W. Define its Kernel and Range. Find the matrix representation of T: P2(R) -> P2(R) defined by T

---

where C_ij is the cofactor of entry a_ij in matrix A.
- The summation sum_{i=1}^n b_i * C_ij represents the cofactor expansion along column j of a new matrix A_j formed by replacing the j-th column of A with the constant column vector B: sum_{i=1}^n b_i * C_ij = det(A_j).
- Therefore: x_j = det(A_j) / det(A) for each j = 1, 2, ..., n. (Proved).
2. Step-by-Step Numerical Solution:
- System:
2x - y + 3z = 9
x + y + z = 6
x - y + z = 2
- Step (a): Compute Delta = det(A):
| 2 -1 3 |
| 1 1 1 | = 2(1 - (-1)) - (-1)(1 - 1) + 3(-1 - 1) = 2(2) + 1(0) + 3(-2) = 4 - 6 = -2.
Since Delta = -2 != 0, Cramer's rule is applicable.
- Step (b): Compute Delta_x = det(A_1) by replacing column 1 with [9, 6, 2]^T:
| 9 -1 3 |
| 6 1 1 | = 9(1 - (-1)) - (-1)(6 - 2) + 3(-6 - 2) = 9(2) + 1(4) + 3(-8) = 18 + 4 - 24 = -2.
=> x = Delta_x / Delta = (-2) / (-2) = 1.
- Step (c): Compute Delta_y = det(A_2) by replacing column 2 with [9, 6, 2]^T:
| 2 9 3 |
| 1 6 1 | = 2(6 - 2) - 9(1 - 1) + 3(2 - 6) = 2(4) - 0 + 3(-4) = 8 - 12 = -4.
=> y = Delta_y / Delta = (-4) / (-2) = 2.
- Step (d): Compute Delta_z = det(A_3) by replacing column 3 with [9, 6, 2]^T:
| 2 -1 9 |
| 1 1 6 | = 2(2 - (-6)) - (-1)(2 - 6) + 9(-1 - 1) = 2(8) + 1(-4) + 9(-2) = 16 - 4 - 18 = -6.
=> z = Delta_z / Delta = (-6) / (-2) = 3.
Final Solution:
x = 1, y = 2, z = 3.
Verification:
2(1) - 2 + 3(3) = 2 - 2 + 9 = 9 (Correct).
1 + 2 + 3 = 6 (Correct).
1 - 2 + 3 = 2 (Correct).

>> Solution to Question 25: Explain LU Decomposition (Doolittle factorization) of a square matrix. Factorize matrix A = [[2, 1, 1], [4, 1, 0], [-2, 2, 1]] into L and U, and

u22 = a22 - l21*u12 = 1 - 2(1) = -1.
u23 = a23 - l21*u13 = 0 - 2(1) = -2.
- Col 2 of L:
l32 = (a32 - l31*u12) / u22 = (2 - (-1)(1)) / (-1) = (2 + 1) / (-1) = -3.
- Row 3 of U:
u33 = a33 - l31*u13 - l32*u23 = 1 - (-1)(1) - (-3)(-2) = 1 + 1 - 6 = -4.
Matrices L and U:
L = [1 0 0] U = [2 1 1]
[2 1 0] [0 -1 -2]
[-1 -3 1] [0 0 -4]
3. Solving AX = B = [1, -2, 7]^T:
- Step (a): Solve L*Y = B:
[1 0 0][y1] [1]
[2 1 0][y2] = [-2]
[-1 -3 1][y3] [7]
y1 = 1.
2*y1 + y2 = -2 => 2(1) + y2 = -2 => y2 = -4.
-y1 - 3*y2 + y3 = 7 => -1 - 3(-4) + y3 = 7 => -1 + 12 + y3 = 7 => 11 + y3 = 7 => y3 = -4.
Vector Y = [1, -4, -4]^T.
- Step (b): Solve U*X = Y:
[2 1 1][x1] [1]
[0 -1 -2][x2] = [-4]
[0 0 -4][x3] [-4]
-4*x3 = -4 => x3 = 1.
-x2 - 2*x3 = -4 => -x2 - 2(1) = -4 => -x2 = -2 => x2 = 2.
2*x1 + x2 + x3 = 1 => 2*x1 + 2 + 1 = 1 => 2*x1 = -2 => x1 = -1.
Final Answer:
X = [-1, 2, 1]^T.

Solution to Question 26: Explain the mathematical theory of Singular Value Decomposition (SVD). Define singular values and express the pseudo-inverse (Moo

- The left singular vectors u_i are defined by:
u_i = (1 / sigma_i) * A * v_i for i = 1, 2, ..., r.
- These satisfy (A * A^T) * u_i = sigma_i^2 * u_i; hence u_i are orthonormal eigenvectors of A * A^T.
3. Dyadic (Rank-One) Form of SVD:
A = sum_{i=1}^r sigma_i * u_i * v_i^T.
4. Moore-Penrose Pseudo-Inverse A^+:
- If A is not square or singular, its Moore-Penrose pseudo-inverse A^+ is given by:
A^+ = V * Sigma^+ * U^T,
where Sigma^+ is an n x m matrix formed by taking the reciprocal of non-zero singular values:
Sigma^+ = [diag(1/sigma_1, 1/sigma_2, ..., 1/sigma_r), 0; 0, 0].
- A^+ solves the general least-squares problem: for any AX = B, the minimum norm least-squares solution is X = A^+ * B.

>>Solution to Question 27: Describe the Power Method algorithm for computing the dominant eigenvalue and eigenvector of a matrix. Apply the method for 4 iterat

2.2361) / 2 = 4.6180.

- The computed value c4 = 4.5067 is rapidly converging to 4.6180 within only 4 iterations.

---

where S in M_{m x n}(R) is the stoichiometric matrix (S_ij = stoichiometric coefficient of metabolite i in reaction j), and mu is the cell growth dilution rate.
- At metabolic quasi-steady state (reaction rates are orders of magnitude faster than cell division rates):
dC / dt ~ 0 => S * v = 0.
2. Null Space Analysis and Degrees of Freedom:
- S * v = 0 is a homogeneous linear algebraic system.
- The feasible flux distributions lie entirely in the Null Space of the stoichiometric matrix:
v in Null(S) = Ker(S).
- By Rank-Nullity Theorem:
Degrees of Freedom (DOF) = dim(Null(S)) = n - rank(S).
- If DOF = 0: All fluxes are strictly uniquely determined.
- If DOF > 0: The system is underdetermined. To determine internal intracellular fluxes, exactly DOF independent external flux measurements (e.g., glucose uptake rate, oxygen consumption, lactate secretion) must be supplied.
3. Worked Illustrative Example:
- Consider 3 metabolites (A, B, C) and 4 fluxes:
R1: Substrate -> A (v1)
R2: A -> B (v2)
R3: A -> C (v3)
R4: B + C -> Cell (v4)
- Mass balances:
For A: v1 - v2 - v3 = 0
For B: v2 - v4 = 0
For C: v3 - v4 = 0
- Stoichiometric Matrix S:
[1 -1 -1 0]
S = [0 1 0 -1]
[0 0 1 -1]
- Row reduction:
Rank(S) = 3 (3 independent rows).
Variables n = 4.
Degrees of Freedom = 4 - 3 = 1.
- Express fluxes in terms of free variable v4:
From row 3: v3 = v4.
From row 2: v2 = v4.
From row 1: v1 = v2 + v3 = v4 + v4 = 2*v4.
- General Solution Vector:
v = [v1, v2, v3, v4]^T = v4 * [2, 1, 1, 1]^T.
- Biological Insight: Measuring a single external uptake flux v1 = 10 mmol/(g*h) immediately determines all intracellular rates: v4 = 5, v2 = 5, v3 = 5 mmol/(g*h).

>> Solution to Question 30: Model a multi-component bioprocess separation column using a linear system of differential equations dX/dt = A*X. Solve the system u

exp(A * t) = P * exp(D * t) * P^-1 = P * diag(e^(lambda_1*t), ..., e^(lambda_n*t)) * P^-1.
2. Step-by-Step Solution for A = [[-2, 1], [0, -3]]:
Step (a): Eigenvalues:
Since A is upper triangular, its eigenvalues are the diagonal entries:
lambda1 = -2, lambda2 = -3.
Step (b): Eigenvectors:
- For lambda1 = -2:
(A - (-2)*I)X = 0 => [0 1; 0 -1][x1; x2] = [0; 0] => x2 = 0, x1 free.
X1 = [1, 0]^T.
- For lambda2 = -3:
(A - (-3)*I)X = 0 => [1 1; 0 0][x1; x2] = [0; 0] => x1 + x2 = 0 => x1 = -x2.
X2 = [-1, 1]^T.
Step (c): Modal Matrix P and Inverse P^-1:
P = [1 -1]
[0 1]
det(P) = 1.
P^-1 = [1 1]
[0 1].
Step (d): Compute exp(A * t):
exp(D * t) = [e^(-2t) 0]
[0 e^(-3t)]
exp(A * t) = P * exp(D * t) * P^-1
= [1 -1] * [e^(-2t) 0] * [1 1]
[0 1] [0 e^(-3t)] [0 1]
= [e^(-2t) -e^(-3t)] * [1 1]
[0 e^(-3t)] [0 1]
= [e^(-2t) e^(-2t) - e^(-3t)]
[0 e^(-3t)].
Step (e): Apply Initial Conditions X(0) = [100, 50]^T:
X(t) = exp(A * t) * X(0)
= [e^(-2t) e^(-2t) - e^(-3t)] * [100]
[0 e^(-3t)] [50]
x1(t) = 100 * e^(-2t) + 50 * (e^(-2t) - e^(-3t)) = 150 * e^(-2t) - 50 * e^(-3t).
x2(t) = 0 * 100 + 50 * e^(-3t) = 50 * e^(-3t).
Conclusion:
The concentrations of the two separating biotechnology components over time t are:
x1(t) = 150*e^(-2t) - 50*e^(-3t)
x2(t) = 50*e^(-3t).
As t -> infinity, both concentrations decay stably to zero.
